

P versus NP: What Random 3-SAT Experiments Can and Cannot Show

Phase transition, solver scaling, a physics-inspired analog solver and learned satisfiability, with PINNeAPPLE

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Abstract

$P \neq NP$ is a statement about worst-case complexity over all instances; no finite experiment on random instances can prove or refute it. We measure what can be measured and state its limits. (1) With a complete CDCL solver (Glucose 4), the 50 % satisfiability crossing of random 3-SAT moves from 4.39 ($N = 50$) to 4.25 ($N = 200$), consistent with the known threshold $\alpha_c \approx 4.267$; the median solver cost peaks at $\alpha = 4.25$ – 4.5 on the 0.25-spaced grid, i.e. at the transition. (2) At $\alpha = 4.26$ the median conflict count grows from 46 ($N = 50$) to 4078922 ($N = 350$); by AIC the exponential fit is preferred over this range. (3) A continuous-time analog solver solved 100% of satisfiable instances within its time budget for $N = 20$ – 80 . (4) A validation-selected PINNeAPPLE classifier on instance statistics reached 85.1% test accuracy against 84.7% for α alone.

Keywords: P vs NP; random 3-SAT; phase transition; CDCL; analog computing; PINNeAPPLE; Millennium Prize problems

What this study shows — and what it does not

- Shows: the average-case behaviour of one solver family on one random ensemble, measured with fixed seeds and reported with its spread.
- Does not bear on P vs NP: random 3-SAT near threshold may be hard on average for reasons unrelated to worst-case complexity, and a polynomial algorithm could exist that no current solver implements. A fit over $N \leq 350$ cannot distinguish exponential growth from a high-degree polynomial with confidence.
- Learning satisfiability from cheap statistics exploits finite- N fluctuations of the ensemble; it is not a decision procedure.

1 Background

Cook and Levin showed that SAT is NP-complete [1, 2]; a polynomial algorithm for 3-SAT would give $P = NP$. For random 3-SAT with $M = \alpha N$ clauses, satisfiability drops sharply near $\alpha_c \approx 4.267$ as $N \rightarrow \infty$ [3, 4], and complete solvers are slowest there [5]. Continuous-time dynamical systems whose attractors are the solutions have been proposed as physics-inspired solvers [6]; their time-to-solution was reported to grow polynomially in analog time at the cost of exponentially growing auxiliary variables.

2 Methods

- Instances: uniform random 3-SAT (3 distinct variables per clause, random signs), fixed seed. Phase transition: $N \in \{50, 100, 150, 200\}$, α from 3.0 to 5.5 in steps of 0.25, 100 instances per point; 50 % crossing by linear interpolation.
- Scaling: $\alpha = 4.26$, 60 instances per N ; median and 90th percentile of CDCL conflicts; exponential ($\log y = a + bN$) and power-law ($\log y = a + k \log N$) fits compared by AIC, both with two parameters.
- Analog solver: the Ercsey-Ravasz–Toroczkai dynamics (soft spins $s \in [-1, 1]^N$, exponentially growing clause weights), explicit Euler with a step cap, budget $t \leq 500$, 30 satisfiable instances per N at $\alpha = 4.25$ (unsatisfiable ones are discarded, since the solver can only succeed on satisfiable instances).
- Learned satisfiability: $N = 100$, $\alpha \sim U(3.8, 4.8)$, 3000 instances labelled by the complete solver; features α , occurrence std and max, literal-bias mean and std, fraction of unused variables; 70/15/15 split; the first

attempt used one fixed PINNeAPPLE ModifiedMLP, the second selects among PINNeAPPLE classifiers on validation only; test set used once.

3 Results

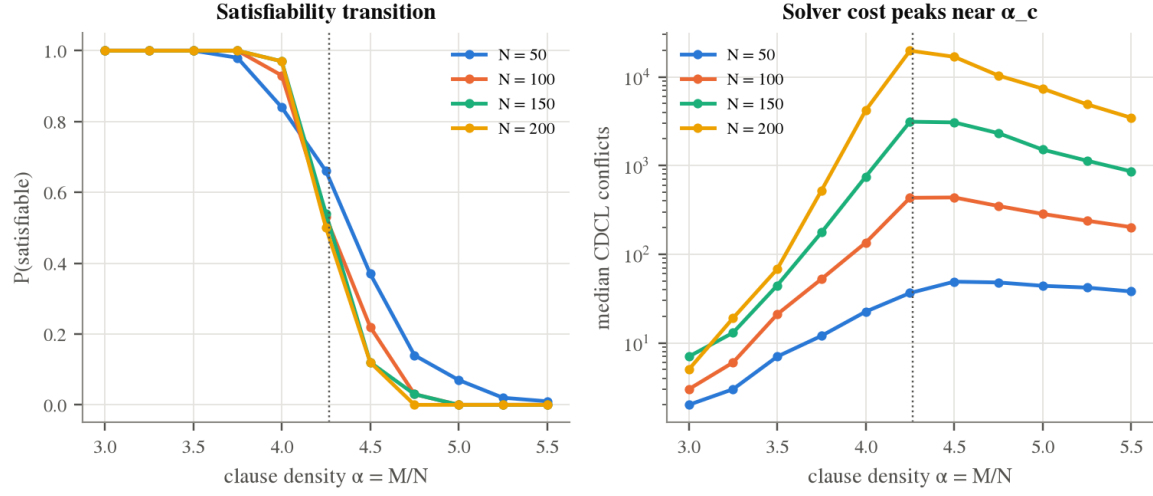


Figure 1. Left: probability of satisfiability; right: median CDCL conflicts. Dotted line: $\alpha_c \approx 4.267$.

Table 1. Finite-size crossing points.

N	α at $P(\text{sat}) = 0.5$	distance to α_c
50	4.388	+0.121
100	4.274	+0.007
150	4.274	+0.007
200	4.250	-0.017

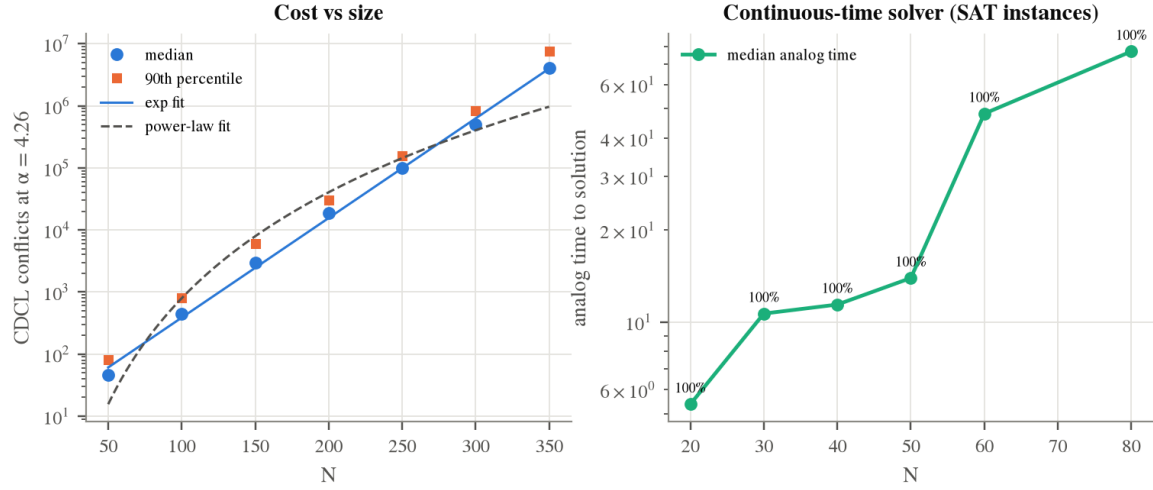


Figure 2. Left: cost at $\alpha = 4.26$ with both fits. Right: analog time to solution; labels give the fraction solved within budget.

Table 2. Fits of median conflicts vs N at $\alpha = 4.26$.

model	coefficients	RSS (log scale)	AIC
exponential	$b = 0.0370$ per variable	0.229	-19.95
power law	$k = 5.67$	5.314	2.07

Both models have two parameters, so the AIC difference is the log-likelihood difference. Over a factor of 7 in N the two curves remain close; the preference is a statement about this range only.

Table 3. Analog solver on satisfiable instances at $\alpha = 4.25$.

N	solved within budget	median analog time
20	100%	5.375
30	100%	10.650
40	100%	11.400
50	100%	13.925
60	100%	48.137
80	100%	76.973

Table 4. Predicting satisfiability at $N = 100$.

model	test accuracy	selected on validation
first attempt: alpha only	80.9%	fixed ModifiedMLP
first attempt: all features	49.6%	fixed ModifiedMLP
alpha only	84.7%	fixed ModifiedMLP
all features	85.1%	fixed ModifiedMLP

Instance statistics beyond α change test accuracy by +0.4 percentage points. Near the threshold, α explains most of what cheap statistics can see; the remainder reflects finite-size fluctuations of the ensemble, not structure a polynomial algorithm could exploit.

4 Discussion

The measurements reproduce the known average-case picture: a sharp transition near α_c and a cost peak there. None of this is evidence for or against $P \neq NP$. What PINNeAPLe adds is the reproducible harness (fixed seeds, validation-only model selection, a single test pass) for probes of this kind; the honest conclusion is that the question is not experimental.

References

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